

# Model Lab

Every prior, same slate — 2026-10-04

SERVED PRIOR

**Served (full season)**

PRIORS COMPARED

**11**

HELD-OUT PA

**34,651**

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## Held-Out Scores

Fitted on plate appearances before 2026-08-25, scored on the 34,651 that came after (actual HR rate 3.04%). No outcome in the test window was visible during fitting.

Metrics are per plate appearance; the probabilities lower down are per game.

**Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors.

| Prior                                                                                          | AUC    | Top-decile lift | Brier   | Log loss | Cal. RMSE |
|------------------------------------------------------------------------------------------------|--------|-----------------|---------|----------|-----------|
| <b>Flat league</b><br>League HR rate as the prior for every hitter.                            | 0.5884 | 1.55×           | 0.02946 | 0.13530  | 0.00587   |
| <b>KNN comps</b><br>Prior is the pooled rate of the nearest comparables.                       | 0.5997 | 1.51×           | 0.02943 | 0.13445  | 0.00482   |
| <b>SVR</b><br>Support-vector regression on the same feature matrix.                            | 0.5958 | 1.53×           | 0.02947 | 0.13525  | 0.00592   |
| <b>Random forest</b><br>Forest regression; a learned similarity metric.                        | 0.5953 | 1.52×           | 0.02945 | 0.13476  | 0.00540   |
| <b>Core (KNN+SVR)</b><br>Blend of KNN and SVR, no forest.                                      | 0.5997 | 1.51×           | 0.02943 | 0.13445  | 0.00482   |
| <b>Forest blend</b><br>KNN + SVR + forest, weights fitted out-of-fold.                         | 0.5997 | 1.51×           | 0.02943 | 0.13445  | 0.00482   |
| <b>Linear regression</b><br>Ordinary least squares on the same features.                       | 0.5967 | 1.51×           | 0.02944 | 0.13480  | 0.00427   |
| <b>Logistic regression</b><br>Binomial logistic on the per-PA home run outcome.                | 0.5963 | 1.55×           | 0.02946 | 0.13491  | 0.00633   |
| <b>Neural net (torch)</b><br>Two-layer PyTorch MLP, PA-weighted, trained on the same features. | 0.5970 | 1.53×           | 0.02945 | 0.13473  | 0.00532   |
| <b>XGBoost</b><br>Gradient-boosted trees, depth 3 with subsampling and L2.                     | 0.5962 | 1.55×           | 0.02946 | 0.13490  | 0.00572   |

**Blend weights** (forest variant): KNN 1.00, SVR 0.00, RF 0.00

**Shrinkage K** per prior: league 251, knn 251, svr 251, rf 251, core 251, ensemble 251, linear 251, logistic 251, nn 251, xgb 251 — estimated by method of moments on out-of-fold residuals, not hand-tuned.

# Thin-Sample Subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,763 PA, actual HR rate 2.44%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

| Prior                                                                                          | AUC    | Top-decile lift | Brier   | Log loss | Cal. RMSE |
|------------------------------------------------------------------------------------------------|--------|-----------------|---------|----------|-----------|
| <b>Flat league</b><br>League HR rate as the prior for every hitter.                            | 0.5558 | 1.20×           | 0.02387 | 0.11519  | 0.01017   |
| <b>KNN comps</b><br>Prior is the pooled rate of the nearest comparables.                       | 0.5650 | 1.41×           | 0.02394 | 0.11585  | 0.01501   |
| <b>SVR</b><br>Support-vector regression on the same feature matrix.                            | 0.5788 | 1.31×           | 0.02416 | 0.12070  | 0.01938   |
| <b>Random forest</b><br>Forest regression; a learned similarity metric.                        | 0.5870 | 1.31×           | 0.02409 | 0.11658  | 0.01719   |
| <b>Core (KNN+SVR)</b><br>Blend of KNN and SVR, no forest.                                      | 0.5650 | 1.41×           | 0.02394 | 0.11585  | 0.01501   |
| <b>Forest blend</b><br>KNN + SVR + forest, weights fitted out-of-fold.                         | 0.5650 | 1.41×           | 0.02394 | 0.11585  | 0.01501   |
| <b>Linear regression</b><br>Ordinary least squares on the same features.                       | 0.5744 | 1.20×           | 0.02404 | 0.11802  | 0.01612   |
| <b>Logistic regression</b><br>Binomial logistic on the per-PA home run outcome.                | 0.5751 | 1.31×           | 0.02406 | 0.11742  | 0.01689   |
| <b>Neural net (torch)</b><br>Two-layer PyTorch MLP, PA-weighted, trained on the same features. | 0.5802 | 1.20×           | 0.02400 | 0.11664  | 0.01421   |
| <b>XGBoost</b><br>Gradient-boosted trees, depth 3 with subsampling and L2.                     | 0.5886 | 1.20×           | 0.02416 | 0.11793  | 0.02056   |

# **Slate Agreement**

Served (full season) · 6/6 shared 0/6 hit

Shohei Ohtani  
Fernando Tatis Jr.  
Jackson Merrill  
Jackson Chourio  
Will Smith  
Teoscar Hernández

KNN comps · 6/6 shared 0/6 hit

Shohei Ohtani  
Fernando Tatis Jr.  
Jackson Merrill  
Jackson Chourio  
Teoscar Hernández  
Will Smith

Random forest · 6/6 shared 0/6 hit

Shohei Ohtani  
Fernando Tatis Jr.  
Jackson Merrill  
Jackson Chourio  
Will Smith  
Teoscar Hernández

Forest blend · 6/6 shared 0/6 hit

Shohei Ohtani  
Fernando Tatis Jr.  
Jackson Merrill  
Jackson Chourio  
Teoscar Hernández  
Will Smith

Logistic regression · 6/6 shared 0/6 hit

Flat league · 5/6 shared 0/6 hit

|       |                    |       |
|-------|--------------------|-------|
| 18.3% | Fernando Tatis Jr. | 17.7% |
| 18.0% | Shohei Ohtani      | 16.8% |
| 17.1% | Jackson Merrill    | 15.9% |
| 15.5% | Jackson Chourio    | 15.2% |
| 15.4% | Will Smith         | 15.0% |
| 15.4% | Manny Machado      | 14.6% |

SVR · 4/6 shared 0/6 hit

|       |                    |       |
|-------|--------------------|-------|
| 18.2% | Shohei Ohtani      | 18.5% |
| 18.1% | Fernando Tatis Jr. | 17.9% |
| 17.1% | Jackson Merrill    | 17.2% |
| 15.6% | Jackson Chourio    | 15.8% |
| 15.4% | Manny Machado      | 15.6% |
| 15.2% | Matt Olson         | 15.3% |

Core (KNN+SVR) · 6/6 shared 0/6 hit

|       |                    |       |
|-------|--------------------|-------|
| 18.6% | Shohei Ohtani      | 18.2% |
| 17.7% | Fernando Tatis Jr. | 18.1% |
| 17.4% | Jackson Merrill    | 17.1% |
| 15.9% | Jackson Chourio    | 15.6% |
| 15.7% | Teoscar Hernández  | 15.4% |
| 15.6% | Will Smith         | 15.2% |

Linear regression · 6/6 shared 0/6 hit

|       |                    |       |
|-------|--------------------|-------|
| 18.2% | Shohei Ohtani      | 18.4% |
| 18.1% | Fernando Tatis Jr. | 18.2% |
| 17.1% | Jackson Merrill    | 17.3% |
| 15.6% | Will Smith         | 15.9% |
| 15.4% | Jackson Chourio    | 15.6% |
| 15.2% | Teoscar Hernández  | 15.5% |

|                                     |              |                           |                                                |  |
|-------------------------------------|--------------|---------------------------|------------------------------------------------|--|
| <b>Shohei Ohtani</b>                |              | <b>18.8%</b>              | <b>Neural net (torch)</b> · 6/6 shared 0/6 hit |  |
| <b>Fernando Tatis Jr.</b>           | <b>18.4%</b> | <b>Shohei Ohtani</b>      | <b>18.6%</b>                                   |  |
| <b>Jackson Merrill</b>              | <b>17.7%</b> | <b>Fernando Tatis Jr.</b> | <b>18.3%</b>                                   |  |
| <b>Jackson Chourio</b>              | <b>15.7%</b> | <b>Jackson Merrill</b>    | <b>17.5%</b>                                   |  |
| <b>Will Smith</b>                   | <b>15.7%</b> | <b>Will Smith</b>         | <b>15.8%</b>                                   |  |
| <b>Teoscar Hernández</b>            | <b>15.5%</b> | <b>Jackson Chourio</b>    | <b>15.8%</b>                                   |  |
|                                     |              | <b>Teoscar Hernández</b>  | <b>15.5%</b>                                   |  |
| <b>XGBoost</b> · 5/6 shared 0/6 hit |              |                           |                                                |  |
| <b>Shohei Ohtani</b>                | <b>18.9%</b> |                           |                                                |  |
| <b>Fernando Tatis Jr.</b>           | <b>17.8%</b> |                           |                                                |  |
| <b>Jackson Merrill</b>              | <b>17.6%</b> |                           |                                                |  |
| <b>Jackson Chourio</b>              | <b>16.0%</b> |                           |                                                |  |
| <b>Teoscar Hernández</b>            | <b>15.6%</b> |                           |                                                |  |
| Manny Machado                       | <b>15.5%</b> |                           |                                                |  |

## Hitter by Hitter



| Hitter                                                                      | Served (full season) | Flat league | KNN comps | SVR   | Random forest | Core (KNN+SVR) | Forest blend | Linear regression | Logistic regression |
|-----------------------------------------------------------------------------|----------------------|-------------|-----------|-------|---------------|----------------|--------------|-------------------|---------------------|
| 19. <b>Freddie Freeman</b><br>Los Angeles Dodgers · vs RHP · 16 HR / 661 PA | 11.4%                | 11.3%       | 11.4%     | 11.4% | 10.9%         | 11.4%          | 11.4%        | 11.4%             | 11.2%               |
| 20. <b>Christian Yelich</b><br>Milwaukee Brewers · vs RHP · 12 HR / 503 PA  | 11.4%                | 11.2%       | 11.4%     | 10.9% | 11.2%         | 11.4%          | 11.4%        | 11.6%             | 11.1%               |
| 21. <b>Austin Riley</b><br>Atlanta Braves · vs LHP · 20 HR / 647 PA         | 11.3%                | 10.2%       | 11.2%     | 11.1% | 10.7%         | 11.2%          | 11.2%        | 11.0%             | 11.1%               |
| 22. <b>Xander Bogaerts</b><br>San Diego Padres · vs RHP · 14 HR / 618 PA    | 10.5%                | 10.6%       | 10.4%     | 10.2% | 10.3%         | 10.4%          | 10.4%        | 10.6%             | 10.2%               |
| 23. <b>Ethan Salas</b><br>San Diego Padres · vs RHP · 2 HR / 45 PA          | 9.8%                 | 9.9%        | 9.7%      | 10.2% | 9.6%          | 9.7%           | 9.7%         | 9.7%              | 9.3%                |
| 24. <b>Sean Murphy</b><br>Atlanta Braves · vs LHP · 4 HR / 128 PA           | 9.7%                 | 9.1%        | 9.8%      | 9.1%  | 9.9%          | 9.8%           | 9.8%         | 9.9%              | 9.3%                |
| 25. <b>Dustin Harris</b><br>San Diego Padres · vs RHP · 0 HR / 163 PA       | 9.4%                 | 11.2%       | 9.3%      | 8.0%  | 8.8%          | 9.3%           | 9.3%         | 9.4%              | 9.0%                |
| 26. <b>Michael Harris II</b><br>Atlanta Braves · vs LHP · 27 HR / 638 PA    | 9.3%                 | 9.2%        | 9.2%      | 9.6%  | 9.1%          | 9.2%           | 9.2%         | 9.3%              | 9.7%                |
| 27. <b>Ozzie Albies</b><br>Atlanta Braves · vs LHP · 24 HR / 700 PA         | 9.1%                 | 9.4%        | 8.9%      | 9.6%  | 9.5%          | 8.9%           | 8.9%         | 9.2%              | 9.1%                |
| 28. <b>Mike Yastrzemski</b><br>Atlanta Braves · vs LHP · 12 HR / 417 PA     | 8.3%                 | 8.2%        | 8.4%      | 8.3%  | 8.2%          | 8.4%           | 8.4%         | 8.0%              | 8.0%                |
| 29. <b>Joey Ortiz</b><br>Milwaukee Brewers · vs RHP · 9 HR / 419 PA         | 7.9%                 | 8.8%        | 7.8%      | 7.8%  | 7.9%          | 7.8%           | 7.8%         | 7.8%              | 7.8%                |
| 30. <b>Tommy Edman</b><br>Los Angeles Dodgers · vs RHP · 8 HR / 334 PA      | 6.9%                 | 7.7%        | 7.0%      | 7.3%  | 6.6%          | 7.0%           | 7.0%         | 7.2%              | 7.0%                |
| 31. <b>Jake Cronenworth</b><br>San Diego Padres · vs RHP · 7 HR / 435 PA    | 6.5%                 | 7.9%        | 6.5%      | 6.6%  | 6.6%          | 6.5%           | 6.5%         | 6.7%              | 6.6%                |
| 32. <b>Mauricio Dubón</b><br>Atlanta Braves · vs LHP · 12 HR / 636 PA       | 6.1%                 | 7.5%        | 6.2%      | 6.5%  | 6.4%          | 6.2%           | 6.2%         | 6.1%              | 6.4%                |
| 33. <b>Cooper Pratt</b>                                                     | 6.0%                 | 7.4%        | 5.7%      | 5.1%  | 5.6%          | 5.7%           | 5.7%         | 5.3%              | 5.2%                |

| Hitter                                                                  | Served (full season) | Flat league | KNN comps | SVR  | Random forest | Core (KNN+SVR) | Forest blend | Linear regression | Logistic regression |
|-------------------------------------------------------------------------|----------------------|-------------|-----------|------|---------------|----------------|--------------|-------------------|---------------------|
| Milwaukee Brewers · vs RHP · 3 HR / 270 PA                              |                      |             |           |      |               |                |              |                   |                     |
| 34. <b>Sal Frelick</b><br>Milwaukee Brewers · vs RHP · 4 HR / 384 PA    | 5.9%                 | 7.5%        | 6.1%      | 5.9% | 5.8%          | 6.1%           | 6.1%         | 5.9%              | 5.9%                |
| 35. <b>Luis Lara</b><br>Milwaukee Brewers · vs RHP · 1 HR / 216 PA      | 5.5%                 | 7.1%        | 5.6%      | 4.5% | 4.7%          | 5.6%           | 5.6%         | 4.2%              | 5.0%                |
| 36. <b>David Hamilton</b><br>Milwaukee Brewers · vs RHP · 4 HR / 387 PA | 5.3%                 | 7.0%        | 5.6%      | 5.5% | 5.1%          | 5.6%           | 5.6%         | 5.5%              | 5.4%                |
| 37. <b>Ha-Seong Kim</b><br>Atlanta Braves · vs LHP · 0 HR / 154 PA      | 4.7%                 | 6.0%        | 4.6%      | 3.9% | 3.8%          | 4.6%           | 4.6%         | 4.6%              | 4.4%                |

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context.